MIL-T-20705A
24 April 1970
SUPERSEDING
MIL-T-20705
30 December 1957

MILITARY SPECIFICATION

TEETH, SCARIFIER, BACK-RIP

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers teeth for back-rip scarifiers for tractors equipped with dozer blades which permit penetrating and ripping the ground when the tractor is traveling in reverse gear only. The teeth are suitable for operation in all types of terrain where rugged and high-strength parts are required.

1.2 Classification. Tooth sizes correspond to the size of shank on which they are mounted, as shown in table I. Teeth shall be furnished in the sizes specified:

Table I. Tooth sizes

Tooth size	Shank thickness (Inches)	Shank size	Tractor drawbar pull (Pounds)
1	1-1/2	1, 2, 3	6,300 to 21,000
4	2	4	21,000 to 30,000
5	3	5 and 6	31,000 to 50,000

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 3830

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:14Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Federal

QQ-S-624	- Steel Bar, Alloy, Hot Rolled and Cold Finished, (General Purpose).
QQ-S-634	- Steel, Bar, Carbon, Gold Finished, (Standard Quality).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

Military

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

Federal

FED. TEST METHOD STD. No. 151	- Metals; Test Methods.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes,
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Brat*, Blocking, and Waterproofing; with Appropriate Test Methods.
MS39000	- Shank, Scarifier, Back-Rip, Bulldozer.
MS39126	- Shank, Scarifier, Back-Rip, Angledozer.
	- Tooth, Surface Ripping, Back-Rip.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:14Z
Check the source to verify that this is the current version before use.

indicated, the issue in effect on date of invitation forbids or request for -proposal shall apply.

AMERICAN WELDING SOCIETY

Standard Qualification Procedure (AWS B3.0)

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, Hew York, N. Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N.Y. 10017).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATON, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tarifff Order Section, 1616 P Street NW, Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Description. The back-rip scarifier teeth shall be of the replaceable, positive lock-on type and shall be used on back-rip scarifier shanks which conform to MS39000 and MS39126.

3.2 Material. The working surface of the tooth shall be a chromium-molybdemum (FS 4100-series) or a nickel-chromium-molybdenum

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:14Z
Check the source to verify that this is the current version before use.

(FS 4300-, 8600-, FS 8700-, or FS 9800-series) steel conforming to QQ-S-624. The working surface of the tooth shall be heat treated to a hardness of 4l5 to 495 Brinnel. The body or parts of the tooth other than the working surface shall be of steel conforming to QQ-S-634, FS l040. Certified reports from steel mills for materials specified shall be acceptable as to chemical composition. The tooth shall be be the positive lock-on type and shall be replaceable by one man within 2 minutes without the use of special tools. A locking pin shall be used to fasten the tooth on the shank and it shall be protected from wear and abrasion. With the tooth driven on the shank and firmly seated in the working position, the positive locking pin shall be installed without causing any movement of the tooth relative to the shank.

3.3 Temperature conditions. The teeth shall perform in any ambient temperature from plus 125°F. to minus 65°F.

3.4 Construction. The tooth shall either be welded, forged or cast construction and the dimensions shall conform to MS39333 Tooth, Surface Ripping, Back-Rip. The tooth and component parts shall withstand a pull of 50,000 pounds away from the shank, without pulling off, breaking, or deforming.

3.5 Identification marking. The tooth shall be identified in accordance with MIL-STD-130.

3.6 Finishing and painting. The tooth shall be cleaned and painted in acoordance with MIL-T-704, type C.

3.7 Workmanship. The tooth shall be free from sand, nonmetallic inclusions, segregation, cracks, seams, pipes, laps, fins, scale, sprues, pits, slag, and other harmful extraneous material.

3.7.1 Steel fabrication. Steel used in the fabrication of the tooth shall provide original quality surface finish, and shall be free from kinks and sharp bends. Steel having an eroded surface is not acceptable. The forming of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true, and all sharp edges and burrs shall be removed. Burned surfaces of flame-cut material shall be free of burrs, sharp edges, and slag. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly except where heat

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T08:14Z
Check the source to verify that this is the current version before use.

treatment is required. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.7.2 Locking pin holes. Locking pin holes shall be accurately punched or drilled and shall have the burrs removed. Locking pin holes in structural members subject to variable loads shall be punched or drilled.

3.7.3 Welders and welding.

3.7.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by any of the following listed codes for the type of welding operation to be performed and that such (qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualification of the ASME Boiler and Pressure Vessel Code.

3.7.3.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connectedly the welds are subjected to proof and service loadings.

3.7.4 Machine work. Tolerance and gages for metal fits shall conform to the limitations specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:14Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for. delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level II.

4.3.2 Examination.

4.3.2.1 Critical defects. Samples selected in accordance with 4.3.1 shall be examined for the critical defects specified in 4.4.1. Presence of a critical defect shall be cause for rejection.

4.3.2.2 Major defects. Samples selected in accordance with 4.3.1 shall be examined for the major defects specified in 4.4.l. AQL shall be 2.5 percent defective.

4.3.3 Tests. Upon successful completion of the examination specified in 4.3.2, the teeth shall be subjected to the tests specified in 4.4.2. Failure of any test shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. Teeth shall be examined for the following defects:

Critical

1. Dimensions and tolerances not as specified.
2. Material not as specified.
3. Welding not as specified.

Major

101. Nonmetallic inclusions.
102. Segregation.
103. Cracks.
104. Seams.

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:14Z
Check the source to verify that this is the current version before use.

105. Pipes.
106. Holes not clean cut.
l07. Laps.
108. Fins.
109. Scale.
110. Slag.

4.4.2 Tests.

4.4.2.1 Hardness test. Test the teeth in accordance with method 243 of FED. TEST METHOD STD. No. 151.Take hardness impressions of the tooth at not more than 2-inch intervals along the entire length beginning at the petit shown on MS39333, Tooth; Surface Ripping, Back-Rip. A Brinell hardness not within the limits specified in 3.2 shall constitute failure of this test.

4.4.2.2 Tooth pin test. Install a tooth on a shank using the locking pin apply not less than a 50,000 pound load in an attempt to pull the tooth from the shank. Pulling the tooth from shanks or breakage of the locking pin or other parts shall constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1. Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordante with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

111. Materials and containers not as specified for level A or B. Each incorrect material or each incorrect container shall be considered a defect.
112. Surfaces requiring the application of a contact preservative not coated for level A.
113. Boxes exceeding 250 pounds not modified with skids for level A or B.

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:14Z
Check the source to verify that this is the current version before use.

114. Strapping not zinc coated for level A, and not placed adjacent to the skids for level A or B.
115. Marking illegible, incomplete, or incorrect.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be level A or C, as specified.

5.1.1 Level A. Unpainted surfaces of the teeth requiring the application contact preservative in accordance with MIL-P-116 shall be cleaned, dried, and coated with type P-1 preservative conforming to the applicable specification listed in MIL-P-116.

5.1.2 Level C. The teeth shall be preserved to prevent deterioration from the supplier to the initial destination for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified.

5.2.1 Level A. The teeth shall be packed in close-fitting plywood boxes conforming to PPP-B-601, style A or B, overseas type. Blocking and bracing of the teeth within the boxes shall be in accordance with MIL-STD-1186. Boxes exceeding 250 pounds gross weight shall be modified with skids in accordance with the box specification. Boxes shall be strapped in accordance with the appendix to the box specification, except strapping shall be flat steel, type 1, class B, and on modified boxes the strapping shall be placed adjacent to the skids. When strapping is placed adjacent to the skids, the filler edge cleats should be wide enough so as not to allow strapping to pass over any voids.

5.2.3 Level B. The teeth shall be packed as specified for level A, the boxes shall be domestic type, and strapping shall not be required to be zinc-coated.

5.2.3 Level C. The teeth shall be prepared for shipment in a manner to afford protection against deterioration at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:14Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The teeth are intended f-or use with scarifier assemblies in ripping the ground by bulldozing-type vehicles during the time the vehicle is traveling in reverse or backing. The teeth are suitable for use in all types of terrain encountered in road construction and similar earth-moving work.

6.2 Ordering data. Procurement documents should specify the folLowing:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Level of preservation and level of packing required (see 5.1 and 5.2).

Custodians:

Army - ME
Air Force - 84

User activities:

Army - CE
Navy - YD,MC

Preparing activity:

Army-ME

Project No. 3830-0078

Source: https://assist.dla.mil -- Downloaded: 2026t0-19T08:14Z
Check the source to verify that this is the currentversion before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-20705A - Teeth Scarifier, Back Rip

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:14Z
Check the source to verify that this is the current version before use.